BAEC ENGINE 3

Re-engagement Preparation

Research and Technical Brief

Project: BAEC Trigger Intelligence

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Software implementation accompanying the conceptual working paper
When the Buyer Is Not In-Market: Buyer-Articulated Evaluation Contingencies in B2B Sales Encounters

Prepare the conversation without pretending the buyer has intent.

Abstract

BAEC Engine 3 is the re-engagement preparation layer of BAEC Trigger Intelligence. It begins only after Engine 2 has produced an eligible human-verified correspondence and a person separately requests preparation. Engine 3 binds that request to the exact upstream condition, review, evidence, disclosure decisions, and context; structures four preparation outputs; validates buyer-facing claims against recorded grounding and external-use permissions; and requires human accept, edit, or reject review before the material can be treated as reviewed preparation. It does not decide that contact should occur, infer purchase or switching intent, create an opportunity, change account state, send a message, place a call, or write to a CRM. Stage D implements the deterministic validator. Stage E demonstrates, with prerecorded synthetic proposal fixtures only, that proposals can be constrained to a strict structural envelope and routed through the unchanged validator. Stage E also documents an unresolved limitation: model-supplied semantic annotations are not authoritative descriptions of the prose they label. No live model execution is part of the completed Engine 3 evidence described in this brief.

Source hierarchy and claim status

The conceptual working paper remains the theoretical source of truth. The Research Contract translates that work into software-facing constraints; released Engine 1 and Engine 2 semantics govern the upstream handoffs; the locked Engine 3 domain contract and validator design govern implementation. Software tests demonstrate implementation behavior, not empirical sales validity. The brief uses the same claim-status convention as the Engine 1 and Engine 2 briefs.

Table 1. Claim-status convention used in this brief.

Status	Meaning	Examples in this brief
DEFINITIONAL	Part of the BAEC or CEE construct.	BAEC definition; evaluation activation is distinct from purchase or switching.
PROPOSED	A theoretical proposition or future extension requiring empirical testing.	Whether prepared re-engagement improves response, evaluation, or commercial outcomes.
MANAGERIAL	An account-management interpretation proposed from BAEC information.	Human reassessment before seller action; monitoring and preparation as bounded support.
IMPLEMENTATION	A software choice used to preserve evidence, authority, or auditability.	Preparation Request; grounding bundle; disclosure dispositions; claim ledger; deterministic refusal.

1. Research problem and Engine 3 purpose

The BAEC workflow begins before a buyer is necessarily in an active evaluation. Engine 1 preserves a buyer-articulated condition under which evaluation could become worthwhile. Engine 2 asks whether new evidence may correspond to that condition and records a human correspondence judgment. Engine 3 addresses the next problem: how can a salesperson prepare for a possible re-engagement conversation without converting correspondence into an unsupported claim about the buyer?

Engine 3 research question

> *Can a human-reviewed correspondence record be converted into useful, evidence-grounded re-engagement preparation without treating correspondence as buyer intent, exposing internal monitoring evidence, or authorizing contact?*

Governing principle

> *Verified correspondence can justify preparation for a human conversation. It does not establish that contact should occur, that the buyer is evaluating, or that the buyer intends to purchase.*

Engine 3 is therefore a preparation layer, not an action engine. Its distinctive function is to keep the content of a re-engagement conversation tied to the buyer's earlier condition and to the evidence a human actually reviewed, while preserving a hard stop before seller action.

2. Position in the BAEC workflow

Table 2. Separation of responsibility across the three engines.

Responsibility	Engine 1	Engine 2	Engine 3
Capture and confirm buyer-articulated condition	Yes	No	No
Monitor authorized sources / capture evidence	No	Yes	No
Human correspondence review	No	Yes	No
Request communication preparation	No	No	Yes
Control external reference / disclosure	No	No	Yes
Validate grounded preparation	No	No	Yes
Human accept / edit / reject of preparation	No	No	Yes
Send, call, CRM action, opportunity creation	No	No	No

The operating chain is Remember → Monitor → Recognize → Prepare. Engine 3 owns only Prepare. It reads prior authority; it does not recreate it. Engine 1 remains authoritative for the confirmed condition and currentness. Engine 2 remains authoritative for the human-reviewed correspondence result.

3. Input eligibility and read-only handoff

Engine 3 begins only from an eligible Engine 2 result and only after a separate human Preparation Request. For Version 1, POSSIBLE_CORRESPONDENCE and NO_CORRESPONDENCE are not eligible for buyer-facing preparation.

Table 3. Engine 3 input eligibility.

Requirement	Rule
Correspondence outcome	Must be HUMAN_VERIFIED_CORRESPONDENCE.
Authority	The Engine 2 review must be authoritative under the locked review boundary.
BAEC currentness	The referenced Engine 1 condition must still be current.
Revalidation	BAEC_REVALIDATION_REQUIRED must not be active.
Evidence state	Required evidence may not be retracted, invalidated, or superseded in a way that makes the result stale.
Lineage	Exact upstream lineage must be reconstructible; ambiguous or stale context fails closed.
Human action	A person must explicitly request preparation. Correspondence alone never starts drafting.

The handoff is read-only. Engine 3 does not reclassify a BAEC, recompute the Engine 2 outcome, upgrade a possible correspondence, broaden the buyer-defined condition, or query new evidence sources. Any upstream change that breaks the bound context makes the current preparation context stale.

4. Preparation Request and grounding bundle

Preparation Request

A Preparation Request is a distinct human action that authorizes drafting only. The person selects the eligible correspondence, channel scope, preparation goal, audience context, and related disclosure decisions. Allowed goals are

narrow: check whether the earlier condition is still relevant, ask whether to revisit the prior conversation, ask what has changed, or prepare to learn whether evaluation has actually reopened.

A Preparation Request is not send authorization, call authorization, campaign authorization, meeting authorization, opportunity authorization, or account-state authorization.

Communication Grounding Bundle

The Communication Grounding Bundle is an immutable point-in-time snapshot for one request. It binds the exact buyer statement, confirmed structured condition, Engine 2 review and findings, usable observations and measurements, source metadata, disclosure decisions, preparation goal, and context digest. Buyer-facing grounding is a strict subset of internal grounding.

> *Internal evidence may help a salesperson understand why a conversation could be relevant. That does not automatically make the evidence appropriate to reveal to the buyer.*

5. Disclosure, claim grounding, and deterministic refusal

External Reference Disposition

Monitoring authorization and external communication permission are separate. A source may be legitimate to monitor while still being inappropriate to mention in a buyer-facing message. Every potentially buyer-facing fact receives a human-controlled disposition. UNKNOWN is the default and fails closed for external use.

Table 4. External Reference Disposition.

Disposition	Meaning
DIRECT_REFERENCE_ALLOWED	The fact may be stated externally at the approved specificity.
PARAPHRASE_ALLOWED	The fact may inform external language, but exact wording, detail, or source identity may not be exposed.
INTERNAL_ONLY	The fact may support seller-facing preparation only.
UNKNOWN	Permission is not established; treated as internal-only for buyer-facing preparation.

Claim ledger and refusal rules

Buyer-facing draft segments are structured so that factual assertions and references to the buyer's earlier statement can be checked against exact grounding. Questions may be ungrounded only when they contain no factual premise. Courtesy or transition language may not smuggle in buyer state, urgency, or an event claim.

Table 5. Examples of deterministic refusal boundaries.

If a draft...	Engine 3 response
uses a factual claim without grounding	Refuse before human review.
uses INTERNAL_ONLY or UNKNOWN evidence externally	Refuse.
quotes where only paraphrase is permitted	Refuse.
reveals source identity without permission	Refuse.
asserts current evaluation, purchase/switching intent, or opportunity status	Refuse.
invents a date, threshold, amount, role, event, product, or competitor claim	Refuse.
implies contact or outreach has been authorized	Refuse.
is bound to stale or mismatched context	Refuse or require rebuild from current context.

6. Version 1 preparation outputs

Engine 3 structures four preparation artifacts. They are conversation preparation, not automated engagement.

Table 6. Version 1 outputs.

Output	Purpose and boundary
Internal Why-Now Summary	Seller-only explanation of why the verified correspondence may matter. May use internal grounding but must preserve the no-intent boundary.
Follow-up Email Draft	Buyer-facing re-entry language using only externally permitted grounding. Reopens a conversation; it does not declare buyer state.
Call Opening Draft	Short, natural, question-led opening tied to the earlier condition and permitted context.
Discovery Question Set	Up to five questions to test whether the original condition remains relevant, what changed, and whether evaluation has actually reopened.

7. Human Preparation Review and action boundary

Every output remains subject to human review. The reviewer may ACCEPT_AS_DRAFTED, EDIT, or REJECT. Human-edited language is a new artifact and must be revalidated. Final review attests that factual assertions are supported, externally stated facts are permitted, unsupported buyer-state or intent claims are absent, and internal-only sources or monitoring methods are not exposed.

Meaning of a reviewed artifact

> *A human reviewed this preparation material against the recorded grounding and disclosure context.*

That statement does not mean the buyer should be contacted, was contacted, is evaluating, intends to purchase, or that an opportunity exists.

Hard stop before action

Engine 3 has no authority to send an email or message, create an external mailbox draft, place a call, create a CRM task, enroll a contact in a sequence, schedule a meeting, create or change an opportunity, or change account state. Any future action capability would require a separate approved authority boundary.

8. Stage D deterministic core

Stage D implemented the locked Stage C validator as an offline deterministic domain core. The implementation includes exact vocabularies, immutable domain records, canonical JSON and SHA-256 binding, eligibility and context validation, human-controlled disclosure checks, fact permission separate from source-identity permission, structured DraftSegment validation, claim-to-grounding checks, question-premise checks, buyer-state and intent overclaim refusals, source-instruction isolation, human-edit grounding rules, action-boundary refusals, and exact deterministic reason codes.

Table 7. Stage D completion evidence.

Check	Recorded result
Locked adversarial corpus	64 of 64 cases conform.
Stage D test suite	87 passed; 0 failed.
Additional domain/boundary/canonicalization tests	23 passed within the recorded Stage D verification.
Live model/network execution	None.
Seller action capability	None.

A Stage D pass means only that a preparation case conforms to the locked software rules. It does not establish buyer evaluation, purchase intent, opportunity status, outreach permission, sales effectiveness, or empirical validity of the BAEC construct.

9. Stage E offline proposal evaluation

Stage E added a strict proposal contract and an offline evaluator around the unchanged Stage D authority boundary. All proposals were prerecorded synthetic JSON fixtures. No provider SDK, API key, HTTP request, live model call, or network

access was used. The proposal cannot create or modify upstream eligibility, disclosure decisions, context, artifact audience, origin, or action authority.

Table 8. Stage E prerecorded evaluation distribution.

Outcome	Count
Reaches Human Preparation Review	7
Refused by Stage D	28
Structurally rejected by proposal contract	14
Recorded human edit refused	1
Recorded human edit reaches review	1

The Stage E fixture package contains 49 prerecorded proposals, 2 recorded human edits, and 4 caller-supplied contexts. Recorded completion results were 147 Stage E tests passed, 234 Engine 3 tests passed, 7,031 full-repository tests passed with 2 deselected, and 183 locked Engine 1/Engine 2 public-release tests passed.

Known limitation: E3-E-SEM-001

MODEL_SEMANTIC_ANNOTATION_NOT_AUTHORITATIVE

Stage E does not independently verify that a model-supplied semantic annotation accurately describes the prose it labels. Byte-identical rendered text can receive a different Stage D result when those annotations differ. This is a documented architectural limitation, not a failed software test. Human Preparation Review remains the semantic control over the prose.

Stage F was designed to study this limitation more directly, but the locked Stage F status is design-only: no live-model execution was authorized as part of the completed evidence described here. Engine 3 therefore makes no live-model reliability claim.

10. Synthetic Alder Ridge / Cobalt case

Buyer-defined condition

“If Cobalt raises the price of our imaging service contract by more than 8% at the July renewal, we would put the contract out for review.”

Engine 2 later records human-verified correspondence using corrected synthetic July-renewal evidence with a current usable measurement of 9.00%. The earlier 12.00% figure remains invalidated history. Even after human-verified correspondence, the buyer is not presumed to be actively evaluating.

Example human disclosure choices

Table 9. Example disclosure controls in the synthetic case.

Grounding item	Example disposition
Buyer statement	PARAPHRASE_ALLOWED
More-than-8% threshold	DIRECT_REFERENCE_ALLOWED
July renewal context	DIRECT_REFERENCE_ALLOWED
Exact 9.00% measurement	PARAPHRASE_ALLOWED
Forwarded notice/source path	INTERNAL_ONLY
Invalidated 12.00% figure	INTERNAL_ONLY

Allowed preparation pattern

When we last spoke, you mentioned that a price increase above 8% around the July renewal could make it worth revisiting the contract. I wanted to reconnect because that condition may now be relevant. Has anything changed on your side, and would it be useful to revisit the conversation?

Disallowed pattern

Your contract went up 9%, so I know you're now looking to switch. Let's schedule a demo this week.

The disallowed version converts correspondence into buyer state and switching intent, uses exact evidence beyond the example disclosure permission, and invents urgency.

11. Public product demonstration

The public presentation layer is a standalone synthetic Streamlit walkthrough built for a 60–90 second demonstration. It uses fictional Alder Ridge Clinics / Cobalt Imaging Supply data and prerecorded preparation outputs. It does not read or modify the Engine 1, Engine 2, or Engine 3 research artifacts and is not evidence about BAEC. Its purpose is explanatory: show the four-stage product story without adding action authority.

Table 10. Public demo sequence.

Step	What the viewer sees	Human / system boundary
Remember	The buyer's exact condition, threshold, timing, and potential action.	The prior buyer statement is preserved; nothing about current intent is inferred.
Monitor	A synthetic July renewal notice shows a +9% change against the >8% condition.	The evidence may correspond; it is not yet a conclusion.
Recognize	Buyer statement and new evidence are shown side by side.	The person chooses whether to verify correspondence.
Prepare	Why-now summary, email draft, call opening, and discovery questions.	Drafts are prerecorded in the public demo; nothing is sent and no opportunity is created.

The public demo preserves the boundary strip: "BAEC prepares the conversation. The salesperson decides whether to have it."

12. What Engine 3 does and does not establish

Table 11. Interpretation boundary.

Engine 3 can establish / preserve	Engine 3 does not establish
The exact upstream correspondence selected for preparation.	That the buyer should be contacted.
The human request and disclosure context used for drafting.	That the buyer is currently evaluating.
Whether a structured artifact conforms to the deterministic rules.	Purchase or switching intent.
Which facts and source identities were permitted externally.	Opportunity status or account-state change.
Human accept, edit, or reject review of preparation.	Sales lift, response lift, conversion lift, or buyer benefit.
A reproducible synthetic software-conformance record.	Empirical validation of BAEC theory.

13. Reproducibility, provenance, and limits

Engine 3 was developed as a sequence of locked research and implementation stages. The public record includes the Stage A domain contract, Stage B adversarial synthetic corpus, Stage C validator design/specification and oracle, Stage D implementation report and manifest, Stage E proposal contract/evaluation report and manifest, and Stage F design materials. These artifacts are retained in the BAEC Trigger Intelligence repository under docs/engine3 and the corresponding test and implementation paths.

- Synthetic data only. No real customer, account, pricing, contact, health, or confidential sales data is required by the Engine 3 research package.
- No live-model execution is claimed by the completed Stage E evidence. Stage F design does not change that boundary.
- Software tests are evidence about implementation behavior only. They do not validate the conceptual theory or prove commercial benefit.
- Engine 3 preserves human authority over whether preparation should begin, what may be disclosed, whether a draft is acceptable, and whether any later conversation should occur.

14. Conclusion

Engine 3 completes the preparation side of the BAEC chain. Engine 1 remembers what the buyer said would make evaluation worthwhile. Engine 2 monitors for evidence that may correspond and records a human judgment. Engine 3 converts one eligible human-verified correspondence into bounded, evidence-grounded preparation while refusing unsupported claims, respecting disclosure permissions, and stopping before seller action.

Remember → Monitor → Recognize → Prepare

The contribution of Engine 3 is not to automate the seller's decision. It is to make a possible re-engagement conversation more traceable to what the buyer actually said and to what the organization is actually permitted to use, while preserving uncertainty about the buyer's present state.

Primary implementation sources for this brief

BAEC Engine 3 Re-engagement Preparation Domain Contract; Stage D Deterministic Core Implementation Report; Stage E Offline Prerecorded Proposal Evaluation Report and Implementation Manifest; Stage F Semantic Reliability Experiment Design and claims-boundary materials; BAEC Trigger Intelligence LinkedIn product demo README. If this brief conflicts with the governing source hierarchy, the higher-authority source prevails.